

PolyFlow-NPU Microarchitecture

Polymorphic-Dataflow Fissionable Systolic Array with Hardware Multi-Tenant Isolation (WS || OS)

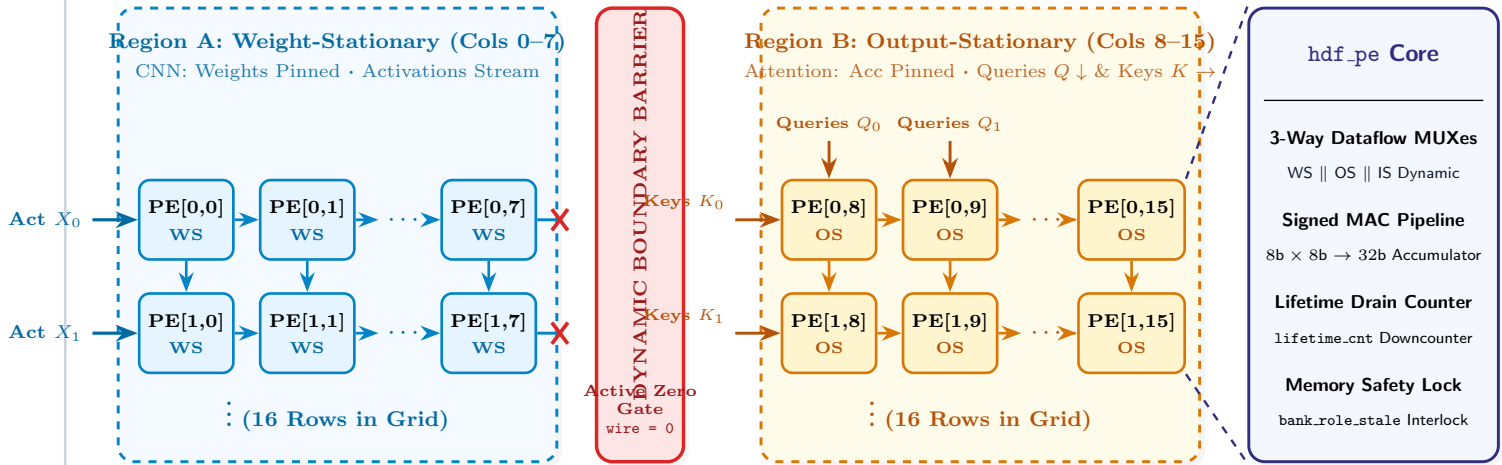
Layer 1: Multi-Tenant Host Scheduler & Dynamic Column Slicing Decoder



Broadcast Mode: Weight-Stationary (WS) Broadcast Mode: Output-Stationary (OS)

Column Split = 8

Layer 2: Heterogeneous Systolic Compute Fabric (16 × 16 Physical PE Mesh)



Region A Memory Bus

Region B Memory Bus

Layer 3: Shared Reconfigurable SRAM Substrate & EPPA Hardware Security Engine

